University of Frontier Technology, Bangladesh
Department of Internet of Things and Robotics Engineering

Project Report

Course Code: IRE-308
Course: System Design, Engineering and Testing Sessional

Submitted by

Md. Mahabub Islam(2201037)
Md. Rahat Hossen(2201028)
Md. Asraful Islam(2201048)
Session:2022-2023
Dept. of IRE
University of Frontier Technology, Bangladesh

Submitted to

Mostafiz Ahammed
Lecturer, Dept. of IRE
University of Frontier Technology, Bangladesh

Date of Submission:31-01-2026.

Project Title: Smart Greenhouse Monitoring System

Abstract

This lab project presents the design and implementation of a Smart Greenhouse Monitoring System using IoT and embedded technology. The system is built around the ESP8266 (NodeMCU) microcontroller, which continuously monitors soil moisture and temperature using appropriate sensors. Based on real-time environmental conditions, the system automatically activates a water pump for irrigation and a cooling fan for temperature regulation. A buzzer alert mechanism is also included to notify users during critical low-moisture conditions. Sensor data is transmitted to an IoT cloud platform for remote monitoring and visualization. The system reduces manual effort, conserves water, and ensures an optimal plant growth environment.

Introduction

Greenhouse farming requires controlled environmental conditions to ensure proper plant growth. Manual monitoring of soil moisture and temperature can be inefficient and may lead to over-irrigation, water wastage, or crop damage. With the advancement of Internet of Things (IoT) and embedded systems, automation has become an effective solution for smart agriculture.

This project introduces an IoT-based smart greenhouse monitoring system capable of real-time sensing, automatic irrigation, temperature regulation, and alert generation. The integration of ESP8266 with cloud monitoring platforms provides remote accessibility and data visualization.

Literature Review

Study / System	Technology Used	Key Features	Limitations
IoT Smart Irrigation Systems	Soil Moisture Sensors, Microcontrollers	Automatic irrigation control	Lack of temperature control
Greenhouse Monitoring Systems	Temperature Sensors, IoT Platforms	Climate monitoring	No integrated irrigation system
Cloud-based Agriculture Monitoring	ThingSpeak, Wi-Fi Modules	Remote data visualization	No alert mechanism

Proposed System	ESP8266, Moisture Sensor, DHT11, Relay, Buzzer	Irrigation + Temperature Control + Alert	Limited to small-scale lab setup

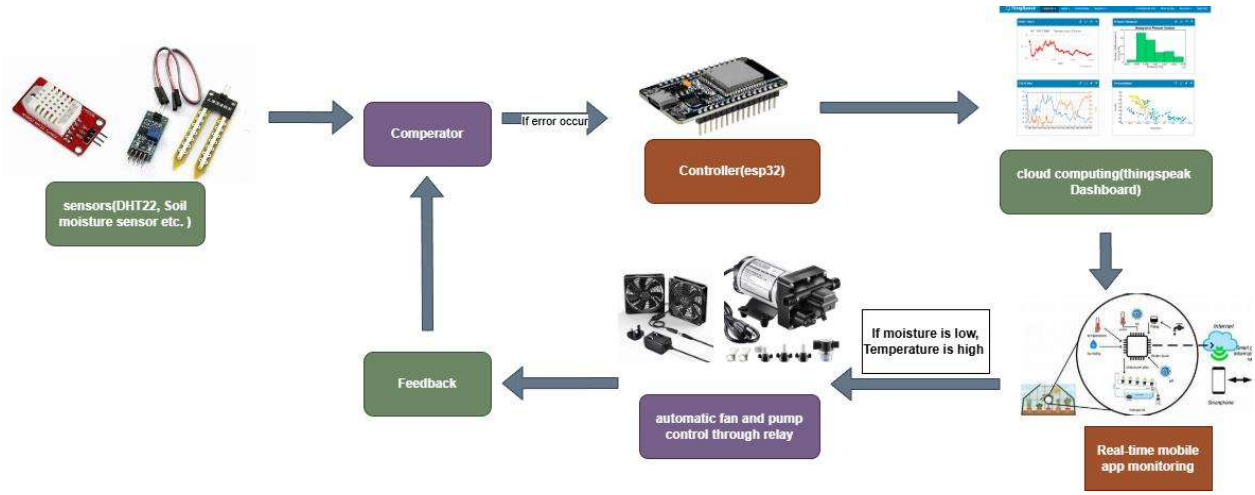
Methodology

The project implementation follows a systematic approach beginning with sensor data acquisition, processing, decision-making, and IoT communication.

First, soil moisture and temperature values are collected using respective sensors. These analog/digital signals are read by the ESP8266 microcontroller. The microcontroller processes sensor data and compares it with predefined threshold values.

If soil moisture falls below the threshold, the system activates the water pump via a relay module to irrigate the plants. Simultaneously, a buzzer alert is triggered. When moisture returns to normal, both pump and buzzer are turned off. If the temperature rises above the safe level, the cooling fan is activated.

All sensor readings are transmitted to the IoT cloud platform for real-time monitoring and analysis.

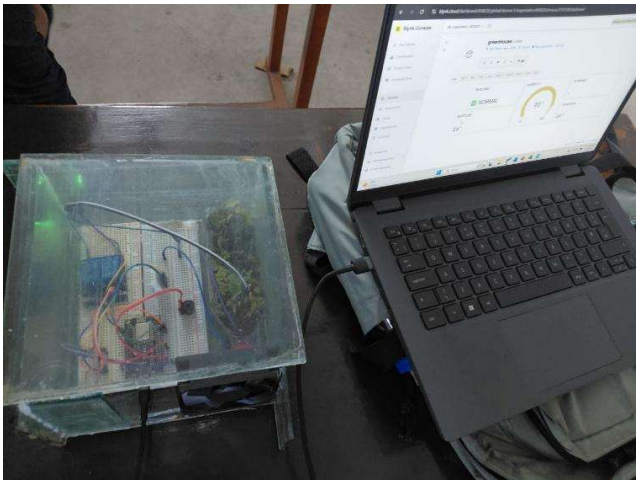


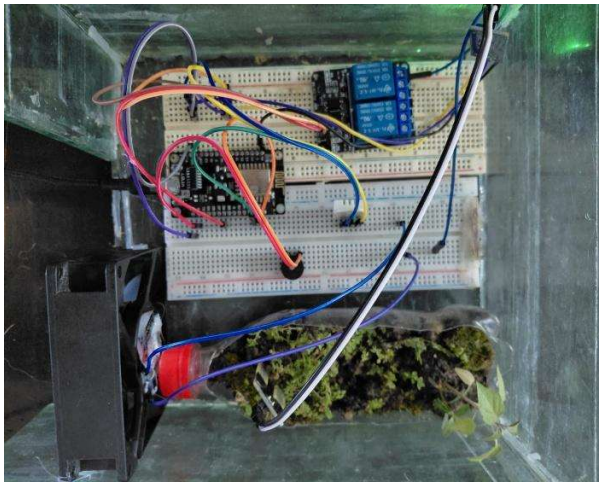
System Design / Implementation

The hardware system consists of ESP8266 (NodeMCU), soil moisture sensor, DHT11 sensor, relay modules, water pump, cooling fan, and buzzer. The ESP8266 acts as the central controller, receiving sensor inputs and controlling output devices.

The system architecture follows a sensor–controller–actuator model. Sensors provide environmental data to ESP8266. Based on decision logic, relays drive the pump and fan, while the buzzer provides audible alerts. Wi-Fi connectivity enables cloud communication.

Project Sample



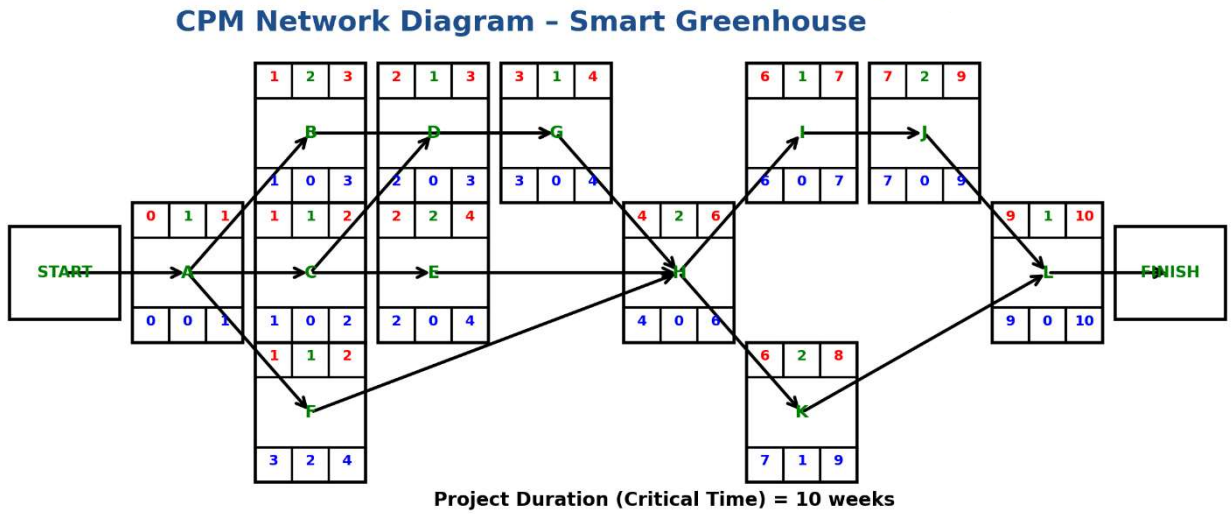


Activity Table

ID	Activity	Predecessor (IDs only)	Duration (Weeks)
A	Requirement & Scope Finalization	—	1
B	Component Selection & Procurement	A	2
C	System Block Diagram + Pin Mapping + Power Plan	A	1
D	Circuit Design & Prototype Plan	C	1
E	Firmware Skeleton (sensor read + threshold logic)	C	2
F	IoT Platform Setup (Blynk/ThingSpeak)	A	1
G	Hardware Assembly (ESP8266 + sensors + buzzer)	B, D	1
H	Firmware Integration (Wi-Fi + IoT + full logic)	E, F, G	2
I	Calibration & Sensor Validation	H	1
J	Testing & Debugging	I	2
K	Documentation + Final Report + Presentation	H	2
L	Deployment / Demo Setup	J, K	1

CPM Analysis:

1) ES, EF, LS, LF, and Float for each activity:



2) Critical tasks

Activities with Float = 0 are critical:

A, B, C, D, E, G, H, I, J, L

(Non-critical: F has float 2 weeks, K has float 1 week)

3) Critical path

Because there are multiple zero-float branches, you have multiple critical paths (all total 10 weeks):

1. **A → B → G → H → I → J → L**
2. **A → C → D → G → H → I → J → L**
3. **A → C → E → H → I → J → L**

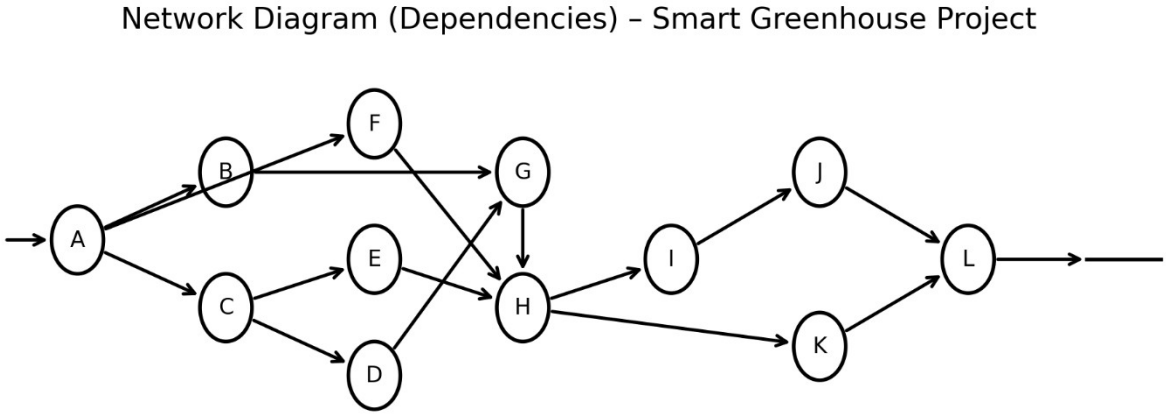
4) Project duration (weeks)

Project Duration = 10 weeks (EF of final activity L).

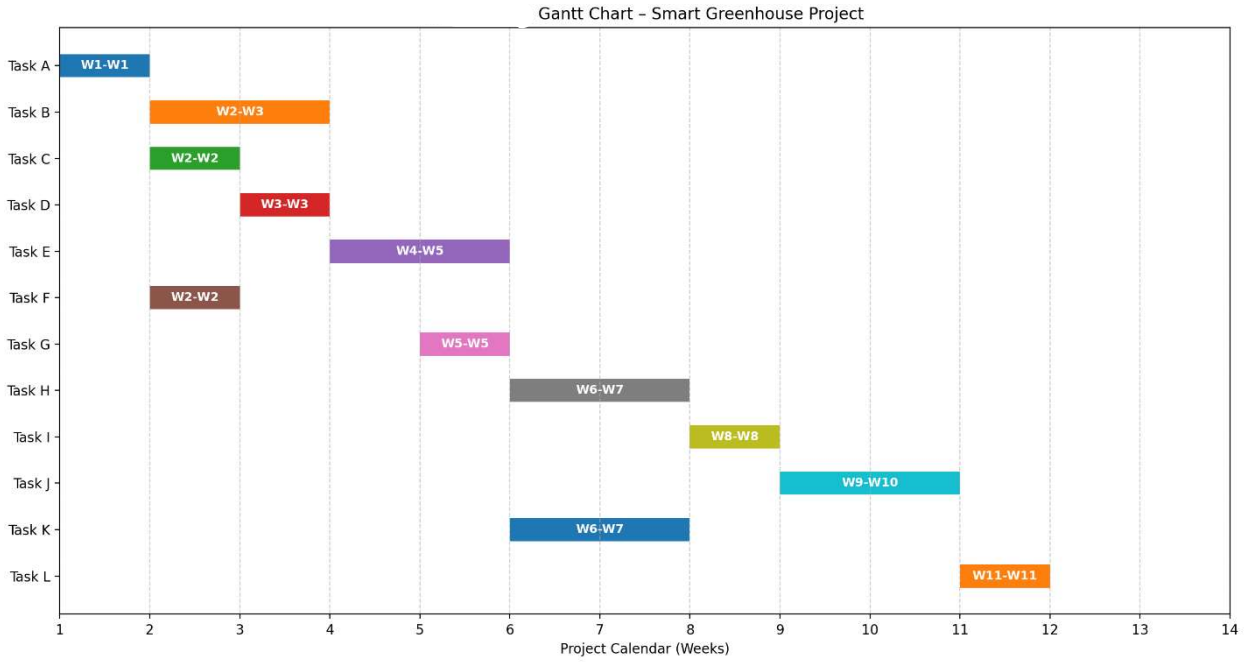
PERT Calculation:

ID	Activity	O	M	P	(T_E=\frac{O+4M+P}{6})
A	Requirement & Scope Finalization	0.5	1	2	1.08
B	Component Selection & Procurement	1	2	4	2.17
C	Block Diagram + Pin/Power Plan	0.5	1	2	1.08
D	Circuit Design & Prototype Plan	0.5	1	2	1.08
E	Firmware Skeleton	1	2	3	2.00
F	IoT Platform Setup	0.5	1	2	1.08
G	Hardware Assembly	0.5	1	2	1.08
H	Firmware Integration (Wi-Fi + IoT + full logic)	1	2	4	2.17
I	Calibration & Validation	0.5	1	2	1.08
J	Testing & Debugging	1	2	4	2.17
K	Documentation + Final Report + Presentation	1	2	3	2.00
L	Deployment / Demo Setup	0.5	1	2	1.08

Network Diagram:



Gantt Chart:



Results and Analysis

The system was tested under different environmental conditions. When soil moisture was low, the pump and buzzer were activated successfully. As moisture increased, the system automatically turned off the pump. Temperature increase resulted in fan activation.

Real-time data monitoring through the IoT platform confirmed accurate sensor readings. The system operated reliably with minimal delay, demonstrating the feasibility of low-cost IoT-based greenhouse automation.

Conclusion and Future Work

This project successfully demonstrates a Smart Greenhouse Monitoring System using IoT and embedded technology. The system automates irrigation, temperature control, and alert generation, making greenhouse management more efficient and resource-friendly.

Future improvements may include humidity control, mobile notification systems, solar-powered operation, and AI-based crop health prediction.

References

IoT-Based Smart Irrigation Systems – IEEE Research Papers

ThingSpeak IoT Cloud Documentation

ESP8266 Technical Reference Manual

Smart Agriculture using Embedded Systems – Journal Articles